Tech Bite #1: How Radio Waves Travel and Behave

What a radio wave actually is, and what happens to it between two antennas -- fields, polarization, wavelength, multipath, and the ionosphere.

Groups T3A + T3B · 24 pool questions · 5-10 minute presentation · Practice exams at eham.net/exams

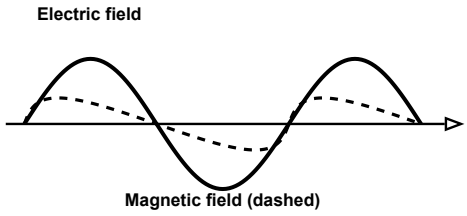
The Big Picture

Radio is invisible, but it is not mysterious – it follows rules you can picture. This first Bite covers two of them: what a radio wave actually is, and what happens to that wave as it travels from one antenna to another. Almost every exam question in this group is just naming an effect once you can see the picture. Get this Bite, and you have the foundation the rest of the exam builds on.

What a Radio Wave Is Made Of

A radio wave is two things traveling together: an **electric field** and a **magnetic field**. Both rise and fall like ripples – in step with each other – and they are locked at right angles, 90 degrees, to each other and to the direction the wave moves. Those two fields are the whole wave; there is nothing else to it.

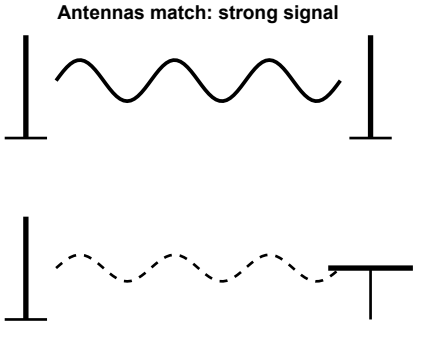
The diagram shows it: one field tracing a sine wave straight up and down, the other tracing the same shape in the plane turned 90 degrees away.



An electromagnetic wave: the electric and magnetic fields oscillate as sine waves, at right angles to each other and to the travel direction.

The two parts of a radio wave are the electric field and the magnetic field, and they sit at right angles to each other. (T3B01, T3B03)

A wave's **polarization** is simply the direction its electric field oscillates. If the electric field goes up and down, the wave is "vertically polarized." (T3B02)



Polarization is a wave's orientation, set by the antenna. Two antennas aligned the same way pass the strongest signal; crossed antennas lose it.

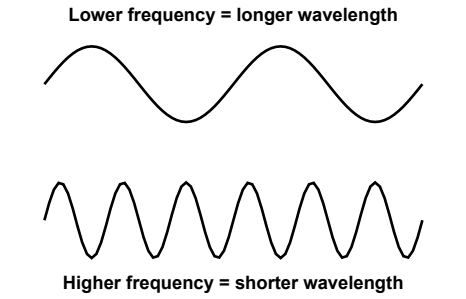
Polarization matters because an antenna works best when it **matches** the wave. A vertical antenna makes and best hears vertically polarized signals; a horizontal antenna favors horizontal ones.

If your antenna is **cross-polarized** – turned 90 degrees from the signal – your **received signal strength drops**. (T3A04) For long-distance, weak-signal VHF and UHF work using CW and SSB, hams agree to use **horizontal** polarization so everyone matches. (T3A03)

Speed, Wavelength, and Why Bands Have Names

Every radio wave travels at the **speed of light** – about **300,000,000 meters per second** – through free space. Microwaves, VHF, UHF: all exactly the same speed. No radio signal is faster or slower than another. (T3B04, T3B11, T3B12)

A wave also has a **wavelength** – the physical length of one full cycle. Wavelength and frequency trade off: **the higher the frequency, the shorter the wavelength.** (T3B05)



Wavelength and frequency always trade off against each other.

Wavelength in meters = 300 divided by the frequency in megahertz (MHz). (T3B06)
Example: a 150 MHz signal is 300 / 150 = 2 meters long. That is exactly why hams call it “the 2-meter band” – bands are named for their approximate wavelength. (T3B07)

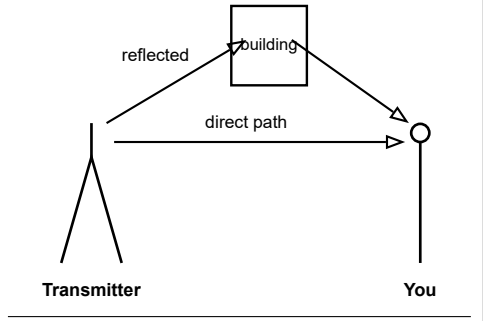
Frequencies are also sorted into ranges, each one ten times the range below it:

Name	Frequency range	Pool questions
HF – High Frequency	3 to 30 MHz	T3B10
VHF – Very High Frequency	30 to 300 MHz	T3B08
UHF – Ultra High Frequency	300 to 3000 MHz	T3B09

What Happens Near You: Multipath

A signal often reaches your antenna by more than one route at once – straight from the transmitter, plus copies that bounced off buildings, hills, or cars. The copies arrive slightly out of step and either **add together or cancel out**. This is called **multipath**.

Reflections are not all bad news. If a building blocks the direct path to a repeater, you can aim a directional antenna to **bounce your signal off another surface** and reach it anyway. (T3A05)



Multipath: one signal reaches you by several paths that add or cancel.

Multipath is why a VHF signal can change strength sharply when you move the antenna just a few feet – you are stepping into a spot where the copies combine differently. (T3A01) In a moving vehicle it causes a rapid flutter called **“picket fencing.”** (T3A06) For digital data, multipath **raises the error rate**. (T3A10)

What Happens Far Away: The Ionosphere and Weather

High above the Earth is a layer called the **ionosphere**. It can **reflect HF radio waves back down to the ground**, letting HF signals travel hundreds or thousands of miles. (T3A11) This is one of the most fun things in the hobby – reaching across the country with modest gear.

Signals that bounce off the ionosphere return **elliptically polarized**, which means **either a vertical or a horizontal antenna will work** to receive them. (T3A09) The ionosphere also causes **irregular fading**, because the signal arrives along several shifting paths that randomly combine. (T3A08)

Weather has small, specific effects worth memorizing:

Condition	Effect	Q
Precipitation (rain, snow)	Decreases range at **microwave** frequencies.	T3A07
Fog or rain on 10 m and 6 m	**Little effect.**	T3A12
Vegetation (trees, leaves)	**Absorbs** UHF and microwave signals, hurting weak-signal reception.	T3A02

Mental Cheat Sheet

- A radio wave = **electric field + magnetic field**, at right angles.
- Polarization = direction of the **electric** field.
- All radio waves travel at the **speed of light**, 300,000,000 m/s.
- Higher frequency = **shorter** wavelength.
- Wavelength in meters = **300 / frequency in MHz**.
- HF 3-30 MHz, VHF 30-300 MHz, UHF 300-3000 MHz.
- Multipath = signal copies adding and canceling; causes picket fencing and more data errors.
- The **ionosphere** reflects HF for long-distance contacts.
- Use **horizontal** polarization for weak-signal VHF/UHF.

Common Traps

- Polarization is set by the **electric** field, not the magnetic field.
- The formula divides **300 by the frequency in MHz** – not the reverse, and not in hertz.
- All radio frequencies travel at the **same** speed. There is no “fastest” one.
- VHF strength changing as you move the antenna is **multipath** – not water vapor, not Doppler.
- Fog and rain barely affect 10 m and 6 m. The “decreases range” answer is for **microwave**.
- HF is reflected by the **ionosphere** – not the troposphere or stratosphere.

Takeaway: A radio wave is an electric and a magnetic field at right angles, traveling at the speed of light. Higher frequency means shorter wavelength, tied together by wavelength = 300 / frequency in MHz. Between antennas, signals bounce: multipath makes nearby signal strength jump around, and the ionosphere reflects HF for long-distance contacts. Match your antenna’s polarization to the signal and you keep the most signal strength.

Bite 1 Question Bank (T3A + T3B)

T3A01. Why do VHF signal strengths sometimes vary greatly when the antenna is moved only a few feet?
A. The signal path encounters different concentrations of water vapor
B. VHF ionospheric propagation is very sensitive to path length
C. Multipath propagation cancels or reinforces signals
D. The Doppler effect causes slight frequency shifts which result in changes in signal strength

T3A02. How does vegetation affect UHF and microwave signals?
A. Causes knife-edge diffraction, distorting voice peaks
B. Absorbs signals, leading to poor reception of weak signals
C. Amplifies signals, improving reception of weak signals
D. Has no effect

T3A03. What antenna polarization is normally used for long-distance CW and SSB contacts on the VHF and UHF bands?
A. Right-hand circular
B. Left-hand circular
C. Horizontal
D. Vertical

T3A04. What is the effect of antenna cross-polarization over a line-of-sight VHF or UHF path?
A. Modulation sidebands might become inverted
B. Received signal strength is reduced
C. Signals have an echo effect
D. Nothing significant will happen

T3A05. When using a directional antenna, how might your station be able to communicate with a distant repeater if buildings or obstructions are blocking the direct line of sight path?
A. Change from vertical to horizontal polarization
B. Try to find a path that reflects signals to the repeater
C. Try the long path
D. Increase the antenna SWR

T3A06. What is the meaning of the term “picket fencing”?
A. Alternating transmissions during a net operation
B. Rapid flutter on mobile signals due to multipath propagation
C. A type of ground system used with vertical antennas
D. Interference from cable TV in the form of carriers at fixed intervals across the band

T3A07. What weather condition might decrease range at microwave frequencies?
A. High winds
B. Low barometric pressure
C. Precipitation
D. Colder temperatures

T3A08. What is a likely cause of irregular fading of signals propagated by the ionosphere?
A. Frequency shift due to Faraday rotation
B. Interference from thunderstorms
C. Intermodulation distortion
D. Random combining of signals arriving via different paths

T3A09. Which of the following results from the fact that signals propagated by the ionosphere are elliptically polarized?
A. Digital modes are unusable
B. Either vertically or horizontally polarized antennas may be used for transmission or reception
C. FM voice is unusable
D. Both the transmitting and receiving antennas must be of the same polarization

T3A10. What effect does multi-path propagation have on data transmissions?
A. Transmission rates must be increased by a factor equal to the number of separate paths observed
B. Transmission rates must be decreased by a factor equal to the number of separate paths observed
C. No significant changes will occur if the signals are transmitted using FM
D. Error rates are likely to increase

T3A11. Which region of the atmosphere can reflect HF radio waves?
A. The stratosphere
B. The troposphere
C. The ionosphere
D. The electrosphere

T3A12. What effect does fog or rain have on 10-meter and 6-meter band signals?
A. Absorption
B. Little effect
C. Deflection
D. Increased range

T3B01. What is the relationship between the electric and magnetic fields of an electromagnetic wave?
A. They travel at different speeds
B. They are in parallel
C. They revolve in opposite directions
D. They are at right angles

T3B02. What property of a radio wave defines its polarization?
A. The orientation of the electric field
B. The orientation of the magnetic field
C. The ratio of the energy in the magnetic field to the energy in the electric field
D. The ratio of the velocity to the wavelength

T3B03. What are the two components of a radio wave?
A. Impedance and reactance
B. Voltage and current
C. Electric and magnetic fields
D. Ionizing and non-ionizing radiation

T3B04. What is the velocity of a radio wave traveling through free space?
A. Speed of light
B. Speed of sound
C. 0.86 times the speed of light
D. 1.86 times the speed of sound

T3B05. What is the relationship between wavelength and frequency?
A. Wavelength gets longer as frequency increases
B. Wavelength gets shorter as frequency increases
C. Wavelength is constant at all frequencies
D. Wavelength and frequency increase as path length increases

T3B06. What is the formula for converting frequency to approximate wavelength in meters?
A. Wavelength in meters equals frequency in hertz multiplied by 300
B. Wavelength in meters equals frequency in hertz divided by 300
C. Wavelength in meters equals frequency in megahertz divided by 300
D. Wavelength in meters equals 300 divided by frequency in megahertz

T3B07. In addition to frequency, which of the following is used to identify amateur radio bands?
A. The approximate wavelength in meters
B. Traditional letter/number designators
C. Channel numbers
D. All these choices are correct

T3B08. What frequency range is referred to as VHF?
A. 30 kHz to 300 kHz
B. 30 MHz to 300 MHz
C. 300 kHz to 3000 kHz
D. 300 MHz to 3000 MHz

T3B09. What frequency range is referred to as UHF?
A. 30 to 300 kHz
B. 30 to 300 MHz
C. 300 to 3000 kHz
D. 300 to 3000 MHz

T3B10. What frequency range is referred to as HF?
A. 300 to 3000 MHz
B. 30 to 300 MHz
C. 3 to 30 MHz
D. 300 to 3000 kHz

T3B11. What is the approximate velocity of a radio wave in free space?
A. 150,000,000 meters per second
B. 300,000,000 meters per second
C. 300,000,000 miles per hour
D. 150,000,000 miles per hour

T3B12. Which of these frequencies travels at the highest velocity in free space?
A. Microwaves
B. UHF
C. VHF
D. All radio frequencies travel at the same velocity

Answer Key

T3A01: C	T3A02: B	T3A03: C	T3A04: B	T3A05: B
T3A06: B	T3A07: C	T3A08: D	T3A09: B	T3A10: D
T3A11: C	T3A12: B	T3B01: D	T3B02: A	T3B03: C
T3B04: A	T3B05: B	T3B06: D	T3B07: A	T3B08: B
T3B09: D	T3B10: C	T3B11: B	T3B12: D	